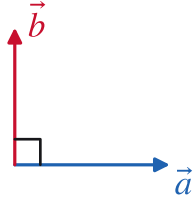
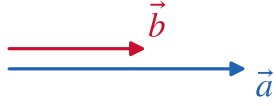


$\alpha = 90^\circ$

$\vec{a} \cdot \vec{b} = 0$



$\alpha = 0^\circ$

$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|$

$\alpha = 180^\circ$

$\vec{a} \cdot \vec{b} = -|\vec{a}||\vec{b}|$